

PhD report for the academic year 2017/2018

Weak solution of the merged mathematical equations of the polluted atmosphere

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Achievements

- Milestone 1
- Milestone 2



Lectures

- 1 Numerical methods for hyperbolic problems - Prof. G. Russo (University of Catania), GSSI
- 2 Physics of the atmosphere - Prof G. Curci, Prof. G. Redaelli, University of L'Aquila



Seminars, workshops, summer schools

- 1 Intensive Program on Fluids and Waves, GSSI
- 2 Women in applied and computational Mathematics, GSSI —
The topic and the results of this present research project were represented - together with the work of many early-stage female researchers - in the poster session.
- 3 Evans function methods in the stability analysis of periodic wavetrains, Prof. R. Plaza
- 4 On the spectral stability of traveling fronts for reaction diffusion-degenerate equations, Prof. R. Plaza
- 5 Waves in Flows Summer School in Prague



University of Sussex - Prof. Omar Lakkis

The main results and steps are the following:

- 1 Software packages such as Docker, Anaconda, ParaView, FEniCS and Alberta have been installed
- 2 Writing scripts to login / connect to the HPC server at the Sussex University
- 3 Getting to know FEniCS: tutorial files and the FEniCS version of the "hello world" programs
- 4 Creating a basis for future work with specific theoretical details that arise when implementing this project - such as the loss of incompressibility when it comes to the FEM approach, and the use of Lagrange multipliers.

